Part II

Upper Extremity

7 The Shoulder *115*

8 The Elbow *129*

9 The Wrist and Hand *140*

II

7 The Shoulder

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Abstract

This chapter describing a systematic approach to magnetic resonance imaging (MRI) of the shoulder takes the reader on a step-by-step review of shoulder MRI, shoulder anatomy, and the most common shoulder pathologic conditions. Shoulder anatomy and its interpretation on MRI can be challenging. The purpose of this chapter is to outline how to perform a detailed and stepwise analysis of shoulder MRI, highlighting both normal anatomy and pathologic conditions. It also describes which sequences are best to assess each structure and pathologic condition. By the end of the chapter, the reader should be able to differentiate normal shoulder anatomy from pathologic conditions on MRI. The ability to evaluate shoulder MRIs is essential for any physician treating patients with shoulder conditions.

Keywords: arthritis, magnetic resonance imaging, shoulder abnormalities, shoulder anatomy, shoulder injuries

7.1 Normal Anatomy

The first section of this chapter describes the anatomic structures of the shoulder and their functions, focusing on how specific structures appear in the sagittal, coronal, and axial planes. This section also introduces an approach to systematically review magnetic resonance imaging (MRI) of the shoulder, assessing individual structures and their relationships to surrounding structures and correlating findings and anatomy between different planes and sequences.

7.1.1 Rotator Cuff

The rotator cuff consists of four muscle-tendon units: the subscapularis, which inserts on the lesser tuberosity and, anteriorly to posteriorly, the supraspinatus, infraspinatus, and teres minor, which all insert on the greater tuberosity. The rotator cuff compresses the humeral head into the glenoid, increasing the lever arm and efficiency of the deltoid muscle. The supraspinatus originates from the supraspinatus fossa, the infraspinatus from the infraspinatus fossa, and the teres minor from the dorsal surface of the scapula and two aponeurotic laminae. The subscapularis originates from the anterior scapular surface. The subscapularis has a tendinous insertion onto the upper two-thirds of the lesser tuberosity and a muscular insertion onto the lower one-third.[1]

It was thought that the supraspinatus footprint was larger than that of the infraspinatus but a recent study demonstrated that the supraspinatus occupies a smaller area of the tuberosity. In an anatomic study, the supraspinatus insertion was triangular with an average medial-to-lateral length of 6.9 mm and an anteroposterior width of 12.6 mm, and the infraspinatus insertion was trapezoidal with a medial-to-lateral length of 10.2 mm and an anteroposterior width of 32.7 mm.[2] Normal rotator cuff tendons are uniformly dark with low signal intensity on all imaging sequences.

Sagittal Anatomy

Starting medially, the sagittal plane shows the rotator cuff muscles at the level of the scapular spine, which is where one assesses for presence and degree of muscle atrophy (▶ Fig. 7.1a). The degree of rotator cuff muscle atrophy is typically graded according to the Goutallier classification.[3] Moving laterally, the musculotendinous junction is visualized near the level of the glenoid. The thick insertion of each rotator cuff tendon is seen laterally. The relative involvement of the rotator cuff tendons in a lateral tear can be determined by following the muscular transitions from their origins medially to their insertions laterally.

Coronal Anatomy

Anteriorly, the subscapularis can be seen inserting laterally onto the lesser tuberosity but it is difficult to assess and characterize on the coronal plane. In contrast, the other rotator muscles can be evaluated for atrophy or fatty infiltration medially, and the tendinous insertions are seen while scrolling anteroposteriorly where they insert on the greater tuberosity. The long head of the biceps tendon (LHBT) splits the greater and lesser tuberosities, with the insertion of the subscapularis seen anteriorly on the lesser tuberosity. The LHBT turns medially as it travels proximal to the transverse humeral ligament and toward its intra-articular insertion. At this level, as the upper border of the subscapularis disappears from view, the rotator interval can be seen laterally between the medially coursing biceps and the far lateral supraspinatus insertion.

Axial Anatomy

Although the insertions of the supraspinatus, infraspinatus, and teres minor can be seen on axial images, they are typically evaluated for abnormal changes on sagittal and coronal images. However, the subscapularis can be well visualized inserting on the lesser tuberosity. The location of the biceps and the bicipital groove are also readily seen in the axial plane. This relationship is associated with assessing for subscapularis tears, as discussed in section “Labrum.” In addition to the location of the biceps, the morphology of the biceps tendon and bicipital groove; signal intensity, including any surrounding fluid; presence of partial or full-thickness tearing; and presence of loose bodies in the sheath or groove can be assessed.[4] A recent study evaluating the accuracy of MRI in detecting abnormality of the LHBT found that MRI is highly sensitive in the detection of full-thickness tears, as well as detecting the absence of complete tear and dislocation or subluxation of the tendon.[5] However, MRI had limited effectiveness in the detection of partial tearing of the LHBT.

7.1.2 Glenoid Labrum and Capsular Ligaments

The labrum is a thick, fibrocartilaginous ringlike structure attached to the glenoid rim. The joint capsule has superior, inferior,

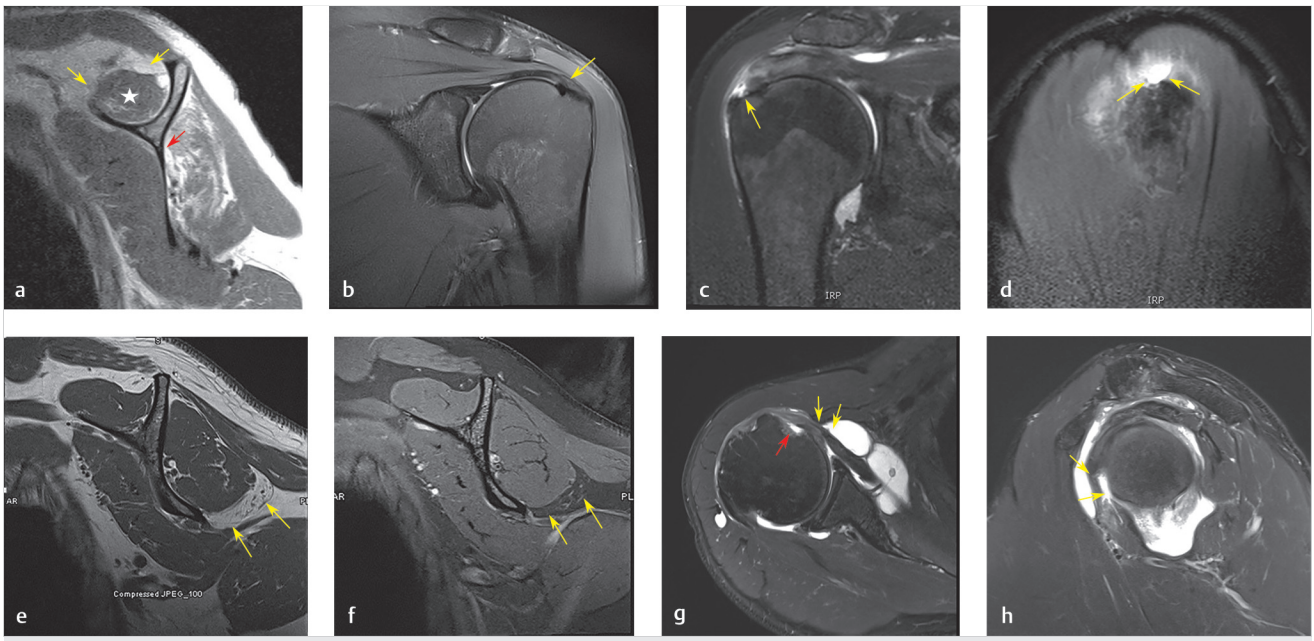


Fig. 7.1 **(a)** Sagittal T1-weighted image at the level of the scapular spine (*red arrow*), with visualization of the supraspinatus muscle belly (*white star*) and surrounding fat atrophy (*yellow arrows*). **(b)** Coronal T2-weighted image of intact supraspinatus insertion at greater tuberosity. **(c)** Coronal T2-weighted image of full-thickness supraspinatus tear with minimal retraction. **(d)** Same supraspinatus tear visualized on sagittal T2-weighted image. Note the size of the tear is more readily apparent as seen between the *yellow arrows*, which mark pathology and relevant anatomic structures. **(e)** Sagittal T1-weighted and **(f)** T2-weighted images with atrophy of the teres minor (*yellow arrows*). **(g)** Axial and **(h)** sagittal images showing subscapularis tear (*yellow arrows*) with noted absence of the biceps from the groove (*red arrow*).

and posterior attachments to the labrum. The labrum gives the otherwise shallow glenoid depth, and the anterior and posterior-inferior labrum provides a stable bumper that contributes to glenohumeral stability. The superior labrum serves as the origin of the LHBT and is a common source of shoulder abnormality (see section "Labrum"), as well as common anatomic variants (see section "Biceps"). Also notable are the superior, middle, and inferior glenohumeral ligaments (IGHLs). The superior glenohumeral ligament, which is located within the rotator interval, restricts anterior humeral head translation with the arm in adduction. The middle glenohumeral ligament, which is seen adjacent and deep to the subscapularis tendon, restricts anterior humeral head translation with the arm in 45 degrees of abduction and external rotation. An anatomic study demonstrated that the superior and middle glenohumeral ligaments attach to the labrum between the 12 o'clock and 1 o'clock positions, with little variability.[6] The IGHL has anterior and posterior bands. The anterior band restricts anteroinferior translation with the arm in 90 degrees of abduction and external rotation, whereas the posterior band restricts posteroinferior translation with the arm in internal rotation and adduction. The IGHL has greater variability in attachment, ranging from the 3 o'clock to the 5 o'clock position.[6]

Sagittal Anatomy

The articular surface of the glenoid and entirety of the glenoid labrum are best seen in the sagittal plane. This view can be helpful in visualizing glenoid bone loss that can occur in the setting of inferior labral tears and instability; however, the coronal and axial planes are more helpful in detecting signal between the glenoid and the labrum in superior labral tear from anterior to posterior (SLAP) and inferior labral tears. The glenohumeral ligaments, including their labral attachments, are typically well visualized in the sagittal plane.

Coronal Anatomy

The superior labrum is best visualized on coronal images (▶ Fig. 7.2a). The biceps tendon can be traced as it courses medially to its intra-articular attachment. Some separation between the superior aspect of the glenoid and the labrum (medial extension of signal) is often normal and correlates with the normal superior labral attachment. It is important to distinguish superior labral tears from normal labral variants, including Buford complex (cordlike middle glenohumeral ligament), sublabral foramen, and sublabral recess.[7] On coronal sequences, SLAP tears are typically posterior to the biceps complex, whereas these anatomic variants will be anterior. Abnormality is suggested when high signal extends laterally into the substance of the superior labrum or when medial signal is present with no underlying cartilage (▶ Fig. 7.2). In addition, SLAP tears will show lateral extension, whereas normal variants will show medial origin and extension.

Axial Anatomy

The anterior and posterior labrum are best visualized on axial images. Typically, T2-weighted sequences will show the labrum to be a dark, triangular-shaped structure on the rim of the glenoid cartilage. The coracoid can be used as a reference for the level (superior to inferior) on the glenoid being viewed. The glenohumeral ligaments can be visualized by scrolling through

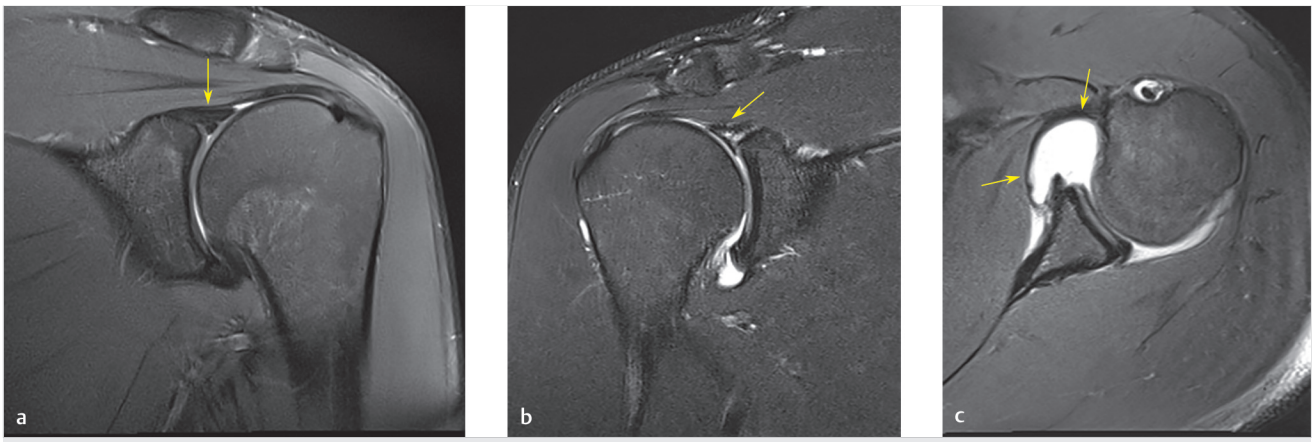


Fig. 7.2 **(a)** Coronal T2-weighted image of intact biceps/superior labral complex (*yellow arrow*). **(b)** Coronal T2-weighted image of torn superior labrum (type 2 superior labral tear from anterior to posterior) (*yellow arrow*). Note lateral extension of signal. **(c)** Superior labral tear seen on axial T2-weighted image (*yellow arrow*).

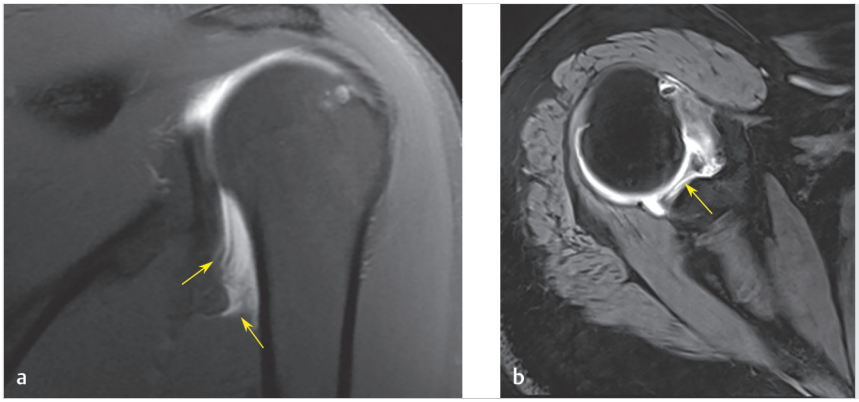


Fig. 7.3 **(a)** Normal humeral attachment of the glenohumeral ligament (*yellow arrows*) seen on axial T2-weighted image. **(b)** Humeral avulsion of the glenohumeral ligament (HAGL) lesion (*yellow arrow*) seen on coronal T2-weighted image.

the images from superior to inferior. At the inferior glenoid in the axillary pouch, the humeral attachment of the glenohumeral ligament can typically be seen attaching to the humeral head (▶ Fig. 7.3a). Superior labral tears, especially larger ones, can also be visualized on axial images.

7.1.3 Biceps

The LHBT originates in the glenohumeral joint from the supraglenoid tubercle and the superior glenoid labrum. A portion of the tendon is intra-articular. There are four variations of the origin of the LHBT. The most common is entirely posterior, followed by posterior dominant, equal, and entirely anterior.[6] As the LHBT traverses toward the bicipital groove, it is held in place by the biceps pulley, which prevents medial subluxation. The biceps pulley consists of the coracohumeral ligament and superior glenohumeral ligament with contributions from the supraspinatus and subscapularis tendons.[8] The biceps travels under the transverse humeral ligament distally.

Sagittal Anatomy

On the sagittal view, the biceps tendon is visualized laterally as it exits the groove adjacent to the lesser tuberosity. It is deep to the supraspinatus tendon and adjacent to the articular surface of the humeral head. The intra-articular portion of the LHBT is seen as it crosses the joint and inserts on the supraglenoid tubercle and the superior glenoid labrum.

Coronal Anatomy

The LHBT is found on the most anterior cuts of the coronal sequences. It appears as a distinct structure and can start to be seen when the tip of the coracoid comes into view. The tendon is seen within the groove and as it turns near the rotator interval, traversing toward its insertion.

Axial Anatomy

The LHBT is easily visualized on axial views and found within the bicipital groove unless subluxed or absent. The bicipital groove separates the greater and lesser tuberosities. The axial view best shows the extra-articular portion of the LHBT.

7.1.4 Glenoid and Scapula

The glenoid is a shallow and concave structure. It is commonly referred to as pear-shaped because of its slightly wider base.

There has been recent interest in using MRI to assess the dimensions (height, width, and depth) of the glenoid and amount of glenoid retroversion as a predictor of glenohumeral instability.[9] Normal glenoid version ranges from neutral to 5 degrees of retroversion.

The scapula is a flat, triangular-shaped bone. It is positioned on the posterior surface of the thoracic rib cage and typically spans ribs 2 to 7. The scapula comprises discrete osseous structures, including the acromion, coracoid process, glenoid, neck, scapular body, and spine. The scapula has the lateral, medial, and superior borders, and the inferior, superior, and lateral angles.

The coracoid process is a critical landmark for arthroscopic examinations and procedures and injections, as well as a critical reference in shoulder MRI review. Attached to the coracoid process are six clinically important structures: the coracoacromial ligament, the coracoclavicular ligaments (conoid and trapezoid), the conjoined tendons of the coracobrachialis and biceps brachii, and the pectoralis minor muscle.

Sagittal Anatomy

Like the glenoid labrum, the articular surface of the bony glenoid is best seen in the sagittal plane. The normal pear shape of the glenoid is seen on this plane, and presence of bone loss (see section "Labrum," shoulder instability) can be assessed. The coracoid and base of the coracoid can be seen while scrolling medially through the images. The coracoid is used as a guide to orient and help differentiate the anterior and posterior aspects of the shoulder. At the level of the scapular spine, the clear separation of the four rotator cuff muscles is seen, and on T1-weighted images at this level, rotator cuff atrophy and fatty infiltration are easily seen and characterized.

Coronal Anatomy

The tip of the coracoid is traced to the glenoid base, and this marks the anterior portion of the scapula and glenoid. Moving posteriorly, the scapular notch is seen and is a landmark for the separation and anterior-to-posterior location of the supraspinatus and infraspinatus. Like the labrum, the superior and inferior aspects of the glenoid are well visualized.

Axial Anatomy

The axial plane offers a clear delineation of the anterior and posterior rim of the glenoid. The coracoid and base of the coracoid should be used as a reference for the reader as the glenoid is assessed. This plane is also helpful to visualize bone loss seen in patients with instability and glenohumeral arthritis.

7.1.5 Humeral Head

The bony anatomy of the proximal humerus and humeral head is well visualized on MRI, and these landmarks serve as references in the review of the shoulder MRI. Important bony landmarks include the anatomic and surgical neck, articular surface, bicipital groove, greater and lesser tuberosities, and humeral shaft. T1-weighted sequences show the bony anatomy well, whereas T2-weighted sequences can further highlight changes within the bone, including cystic change (i.e., in the greater tuberosity) and bony edema. Fat-suppressed three-dimensional gradient-echo sequences are best used for evaluation of articular cartilage.[10]

The humeral head has an ellipsoidal shape with an articular surface of hyaline cartilage. The articular surface of the humeral head approximates one-third of a sphere.[11,12] The anatomic neck of the proximal humerus is the area of transition from cartilage to metaphyseal bone. The humeral head is retroverted relative to the epicondylar axis. Mean humeral head retroversion is approximately 20 degrees, with a range of 15 to 45 degrees.[11]

Sagittal Anatomy

In the sagittal plane, the greater tuberosity with the insertion of the posterior rotator cuff (supraspinatus, infraspinatus, and teres minor) is seen laterally. Cystic changes within the head, adjacent to the rotator cuff insertion, can be seen and indicate chronic rotator cuff disease. Similarly, the subscapularis is seen as it inserts on the lesser tuberosity.

Coronal Anatomy

The coronal plane offers good visualization of the superior and inferior aspects of the humeral head, in addition to visualization of the greater tuberosity and rotator cuff insertions (▶ Fig. 7.1b). Although the lesser tuberosity can be seen, it is better viewed on sagittal and axial images. Coronal images also give the best view of the neck of the humerus and neck-shaft angle, and they allow analysis of superior migration of the humeral head relative to the acromion (see section "Rotator Cuff").

Axial Anatomy

Axial anatomy of the humeral head gives a clear assessment of the greater and lesser tuberosities and the bicipital groove that separates them. The articular cartilage can be examined closely (see Chapter 1, Essentials of MRI Physics and Pulse Sequences) and the version of the humeral head along with any anterior or posterior translation or subluxation of the humeral head relative to the glenoid can be assessed.

7.2 Systematic Approach to Image Review

Just as the physical examination of the shoulder must include all critical structures in a systematic fashion, MRI must be reviewed in a stepwise fashion. To evaluate a shoulder MRI thoroughly, the reader must perform a systematic assessment of the critical structures in all planes and on multiple sequences. If the same steps are performed with every shoulder MRI, the interpreter ensures a full and accurate assessment of all structures and potential abnormalities. The sequence below follows the same pattern as the description of normal anatomy above.

7.2.1 Rotator Cuff

The supraspinatus and infraspinatus are readily seen on coronal and sagittal views. The subscapularis and teres minor can be

seen on coronal images but are more readily seen on axial and sagittal images. A rotator cuff tear that is seen on the coronal T2-weighted image (▶ Fig. 7.1c) should be correlated with the coronal T1-weighted images. To assess the tear size, the reader can count the number of frames (i.e., 3) and multiply by the size of the cuts (3 cuts × 5 mm/cut = 1.5-cm tear). The size of the tear can also be measured directly on the sagittal sequences (▶ Fig. 7.1d). The sagittal view also aids in assessing full-thickness versus partial-thickness tears. The tendons of all four rotator cuff muscles can be followed on the sagittal sequence to their insertions on the tuberosities. In addition, at the level of the scapular spine, the presence and degree of any rotator cuff muscle atrophy can be assessed (▶ Fig. 7.1a). The more severe the grade of atrophy, the larger the tear and the more likely it is chronic.

7.2.2 Glenoid Labrum and Capsular Ligaments

The anterior and posterior labrum are best seen on axial T2-weighted images. Careful review of the axial cuts allows the reader to determine the presence of a tear and at what level on the glenoid the tear starts and ends. On axial views, the normal labrum should appear triangular and wedge-shaped with homogeneously low signal on all sequences.[13] The sagittal view also allows visualization of the entire labral complex. In addition, the sagittal view allows for assessment of any disruption of the superior, middle, or IGHLs.

The superior labrum, like the anterior and posterior labrum, not only can be seen on the axial sequences but also is well visualized on coronal T2-weighted images. The extension of signal medially under the labrum is often normal, representing the normal labral anatomy and attachment to the supraglenoid tubercle.[14] Extension of signal laterally often indicates a superior labral or SLAP tear.[15] Normal variants of labral attachment in the anterior-superior quadrant exist; therefore, potential tears in this area must be interpreted with caution and differentiated from normal anatomic variants.

7.2.3 LHBT

The biceps can be seen on coronal imaging and is a good landmark to evaluate the rotator interval (anterior) and the anterior insertion of the supraspinatus (posterior). It is visualized easily in the bicipital groove on axial sequences. Medial subluxation of the tendon can indicate a subscapularis tear or a tear of the biceps pulley, both of which should be evaluated if medial subluxation is encountered.

7.2.4 Bone and Cartilage Defects

The humeral head should be assessed in all planes. Sagittal and axial views can be used to assess for the presence and size of a Hill-Sachs lesion, which can occur with shoulder instability. The humeral head can also be assessed in the axial plane for any cartilage defects. Cystic changes, which can occur with chronic rotator cuff disease, can be seen in the coronal, sagittal, and axial planes. Changes to the articular cartilage, loss of joint space, and subchondral cystic changes of the glenoid and humeral head can be assessed on axial and coronal cuts. Subchondral bone edema that occurs with degeneration of the glenohumeral joint can best be seen on T2-weighted and gradient-echo sequences.

7.2.5 Acromioclavicular Joint

The acromioclavicular (AC) joint can be seen on axial and coronal series. Edema in the distal clavicle and/or the medial acromion can be assessed in both planes and is best seen on T2-weighted sequences. Associated osteophytes and the presence of subchondral cystic change can be seen on T1- and T2-weighted sequences.

7.3 Traumatic Conditions

Patient age is an important determiner of the most likely shoulder abnormality. In addition, the mechanism of injury is an important factor. Rotator cuff tears are common in patients older than 40 years, whereas shoulder instability and inferior labral tears are most common in patients aged 18 to 35 years. When elderly patient dislocate their shoulders, clinicians must closely evaluate not only the labrum for injury but more commonly, an injury to the rotator cuff, including possible fracture of the greater tuberosity.[16,17]

7.3.1 Labrum

Superior Labral Lesions

Tears of the labrum can occur anywhere in the 360 degrees of labrum that surrounds the glenoid. They can occur in isolation or combination (i.e., an inferior labral tear with a separate superior labral tear). Tears of the anteroinferior and posteroinferior labrum are associated with glenohumeral dislocations and instability, whereas tears of the superior labrum can be degenerative, can occur in throwers, or can result from acute injury. Magnetic resonance arthrography (MRA) has been shown to be more sensitive than noncontrast MRI for assessment of labral and capsular injury.[18]

In a retrospective review of 700 consecutive arthroscopic shoulder procedures, Snyder et al[19] described a tear of the superior labrum in 27 patients. The most common mechanism was fall onto an outstretched arm with the arm in an abducted and slightly forward flexed position. The original classification included the following types: type 1, degeneration and fraying of superior labrum with intact biceps anchor; type 2, detachment of superior labrum and biceps anchor from superior glenoid insertion; type 3, bucket handle labral tear; and type 4, bucket handle labral tear with extension into the biceps tendon (▶ Fig. 7.4h). Combined lesions that do not fit into the above groups have also been described, with types of SLAP tears expanded.[20] SLAP tears can be seen on coronal T2-weighted images (▶ Fig. 7.2b). Seen arthroscopically, the normal labrum can have some liftoff from the cartilaginous superior glenoid rim. Thus, some increased signal under the labrum medially can be normal. Signal that extends laterally on coronal imaging, however, suggests a SLAP tear. In addition, just as inferior tears are well visualized on axial imaging, superior labral tears can sometimes be seen on axial T2-weighted images depending on size and location of tear (▶ Fig. 7.2c). On MRI, SLAP tear appearance depends on the type of tear, as follows[13]: type 1, hyperintense signal with blunting/irregularity of the typically sharp

labral edge; type 2, increased signal intensity and medial extent of signal under the anchor with no signal extension into the biceps tendon, which remains attached to the avulsed labrum; type 3, high signal intensity across the base of the labrum with extension beyond the equator, and if tear is displaced, this can sometimes be seen on axial images with signal interposed between the medial and lateral flap; and type 4, hyperintense signal with extension beyond the equator of the superior labrum with disruption of the superior labrum and extension of signal into the biceps tendon. In addition to acute traumatic superior labral tears, SLAP tears also occur from degeneration and are associated with chondromalacia, glenohumeral arthritis, and rotator cuff tears. In an analysis of 140 consecutive arthroscopic shoulder procedures with superior labral tears, Snyder et al[21] found that 29% of SLAP tears were associated with partial-thickness rotator cuff tears, 11% with full-thickness tears, and 22% with Bankart lesions.

Shoulder Instability

Shoulder instability includes a spectrum of conditions ranging from subluxation to complete dislocation, unidirectional instability (anterior and posterior), and multidirectional instability. MRI and MRA are essential tools in the diagnosis and treatment of shoulder instability.

The most common direction of primary shoulder instability is anterior.[22] Traumatic injuries are the most common cause of shoulder instability and account for 95% of anterior shoulder dislocations. Indirect force with the arm in an abducted and externally rotated position increases the anterior translational forces of the humerus within the glenoid. With the shoulder in this position, the anterior band of the IGHL and the anterior-inferior labrum detach from the glenoid, producing the "essential" lesion of anterior shoulder instability, known as a Bankart lesion.[23] Anteroinferior labral detachment disrupts the main anterior static restraints to the glenohumeral joint in the abducted and externally rotated position, and this must be identified and is often repaired surgically to prevent further instability events. Although a Bankart lesion can often be seen in all planes, it is typically best visualized on axial images with increased linear signal (best seen on T2-weighted imaging) between the labrum and glenoid rim (▶ Fig. 7.4a, b).[24] Tears can be nondisplaced or displaced (typically medially). An anterior labroligamentous periosteal sleeve avulsion occurs when the periosteum remains intact and the labroligamentous structures displace, and commonly scar, medially and rotate inferiorly on the scapular neck. Anterior labroligamentous periosteal sleeve avulsion lesions, like other inferior labral lesions, are best visualized on axial MRI as medial displacement and inferior rotation of a torn anterior labrum (▶ Fig. 7.4c).[24,25] A humeral avulsion of the glenohumeral ligament (HAGL) occurs in less than 10% of instability events but can easily be missed during arthroscopy if not visualized and assessed. On MRI, an HAGL lesion can be seen on axial and coronal images. On axial imaging, HAGL lesion is suggested when a wavy and irregular ligament that does not reach its normal humeral insertion is seen in the axillary pouch. On coronal sequences, the classic appearance is similar to the letter "J," which describes the characteristic wavy appearance of the avulsed ligament from its normal humeral attachment (▶ Fig. 7.3a, b).[26]

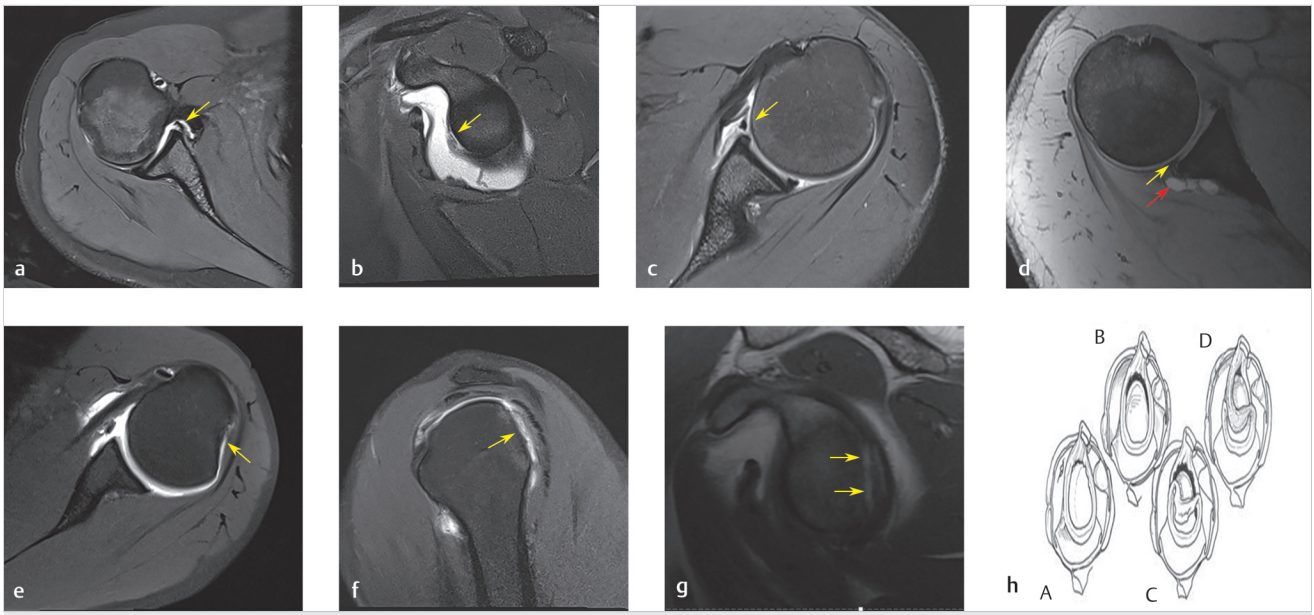


Fig. 7.4 **(a)** Anterior-inferior labral tear (Bankart lesion) (*yellow arrow*) seen on axial T2-weighted image. **(b)** Anterior-inferior labral tear (Bankart lesion) seen on sagittal T2-weighted image. **(c)** Axial T2-weighted image of anterior labral periosteal sleeve avulsion (*yellow arrow*). **(d)** Posterior-inferior labral tear (*yellow arrow*) with associated paralabral cyst (*red arrow*) seen on axial T2-weighted image. **(e)** Hill-Sachs lesion (*yellow arrow*) seen on axial T2-weighted image. **(f)** Same Hill-Sachs lesion (*yellow arrow*) seen on sagittal imaging. **(g)** Reverse bony Bankart lesion (*yellow arrows*) on T1-weighted sagittal image. **(h)** Types of SLAP lesions. A, type I, a frayed and degenerative superior labrum with a normal (stable) biceps tendon anchor. B, type II, detachment of the superior labrum and biceps anchor. C, type III, a bucket-handle tear of a meniscoid superior labrum without extension into the biceps tendon. The biceps anchor is stable and the remaining labrum is intact. D, type IV, a bucket-handle tear of a meniscoid superior labrum with extension into the biceps tendon. (Figure part h: Reprinted from Operative Techniques in Sports Medicine. Vol 12. Powell SE, Nord KD, Ryu RKN. The diagnosis, classification, and treatment of SLAP lesions. 99–110, 2004 with permission from Elsevier.)

In addition to injury to the soft-tissue structures, bony abnormality can also occur. The anteroinferior glenoid may fracture when the humeral head dislocates, creating a bony Bankart lesion. These can be seen as an avulsion or flattening of the anterior-inferior glenoid margin. Glenoid bone loss resulting from recurrent dislocation is important in treatment and predicting recurrence of instability.[27,28] Although computed tomography has historically been the modality of choice for assessment of the amount of glenoid bone loss, MRI is increasingly being used to quantify the degree of bone loss with equal accuracy.[29,30]

Sagittal views allow the best global view and assessment of glenoid bone loss. The Hill-Sachs lesion is the corresponding humeral lesion to the glenoid Bankart lesion. It occurs when the softer bone of the posterosuperior humeral head impacts against the harder anteroinferior glenoid rim. Hill-Sachs lesions are posterosuperior compression fractures on the humeral head and must be recognized because failure to address bony abnormality is associated with future instability episodes. The size of the lesion can be quantified on axial and sagittal images (▶ Fig. 7.4e, f). A glenolabral articular disruption is a defect of the articular cartilage (fibrillation, erosion, delamination) adjacent to a superficial and nondisplaced tear of the anteroinferior labrum.[26] Best seen on axial images, high-intensity signal or contrast is seen adjacent and under but not completely between the glenoid and labrum (▶ Fig. 7.5).

Injury to the middle and inferior one-third of the subscapularis is associated with anterior shoulder dislocation and must be critically assessed. In one study of 45 consecutive patients with anterior shoulder dislocations, more than 60% (27 patients) had some degree of subscapularis injury on MRI.[31]

Posterior instability is uncommon compared with anterior instability and dislocations. The classic mechanisms include seizures or electrical shock in which the more powerful pectoralis muscles produce a posteriorly directed force on the humeral head. More commonly, however, posterior subluxation or dislocation is encountered in athletes such as football linemen. Posterior instability may produce lesions similar to those found in anterior instability and dislocations but on the posterior (or reverse) aspect of the shoulder instead of the anterior aspect. Similarly, the appearance of high-intensity signal or fluid between the posterior labrum and glenoid is characteristic of a tear: a posteroinferior glenoid lesion (reverse Bankart lesion) (▶ Fig. 7.4g) and an anterosuperior humeral head lesion (reverse Hill-Sachs lesion) Flattening and blunting of the normal, sharply contoured labrum is unique to the posterior labrum, whereas the junction between the labrum and glenoid remains intact, which is known as a Kim lesion.[32,33]

Paralabral cysts can be associated with labral tears in any quadrant. The presence of a labral cyst indicates a labral tear because the tear creates a one-way valve that allows the collection of fluid, forming the cyst. Cysts in the spinoglenoid and suprascapular notch are notable because they may cause compression of the suprascapular nerve, which can manifest as pain and/or weakness.

In older patients with shoulder dislocation, the typical mechanism of injury, as well as the pathologic findings, differs from the labral abnormality seen in younger patients.[16] In patients older than 40 years, a low-energy fall is a typical mechanism of injury, and the rotator cuff, not the labrum, is typically torn. In addition, shoulder instability can be caused by massive rotator cuff tearing. Associated bony injuries can also occur, especially in elderly patients with osteoporosis. These include impaction of the humeral head, greater tuberosity fracture, and anterior glenoid rim fracture (▶ Fig. 7.6).

7.4 Degenerative Conditions

7.4.1 Rotator Cuff: Tendinopathy, Partial- and Full-Thickness Tears

Rotator cuff disease is highly correlated with age, with 28% of patients with partial- or full-thickness tears older than 60 years and 65% of patients older than 70 years.[34] In addition, tears are bilateral in 50% of patients older than 66 years.[34]

Degeneration and inflammation of the rotator cuff can lead to rotator cuff tendinopathy. Tendinopathy can lead to partial- and full-thickness tearing. However, the tendon degeneration that occurs without actual tendon tearing is irreversible. There are different theories about the pathogenesis and cause of rotator cuff tendinosis. Neer[35] described extrinsic factors that put a critical portion of the supraspinatus at risk, including variations in acromial morphology and shape, osteophyte formation, and the acromial attachment of the coracoacromial ligament. Neer[35] hypothesized that most rotator cuff tears (95%) were the result of this rotator cuff impingement. Acromioplasty of this bone was developed in an attempt to prevent impingement and subsequent rotator cuff tearing. The shape of the undersurface of

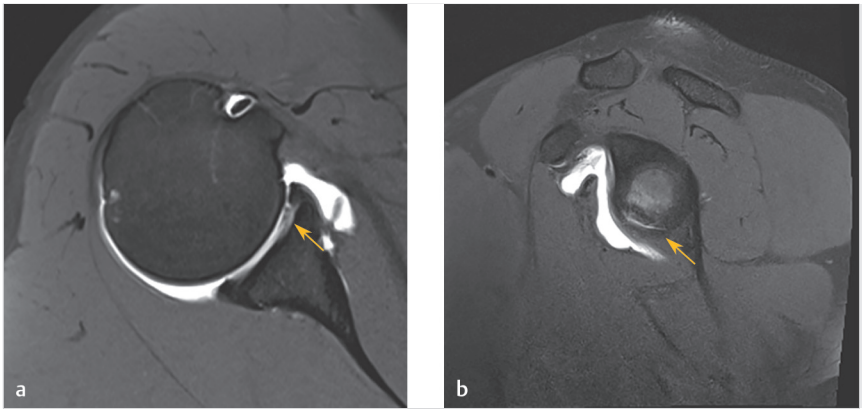


Fig. 7.5 Glenolabral articular disruption lesion (*yellow arrows*) on **(a)** axial and **(b)** sagittal T2-weighted images.

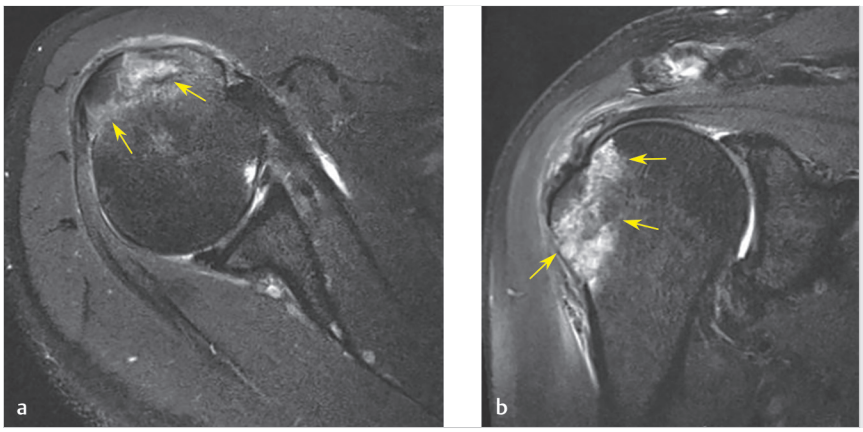


Fig. 7.6 Displaced greater tuberosity fracture (*yellow arrows*) seen on **(a)** axial and **(b)** coronal T2-weighted images.

the acromion is classified into three distinct types: type 1, flat; type 2, curved (smooth); type 3, hooked.[36] Acromial shape in the anterior-to-posterior plane is best visualized on scapular lateral radiographs and, correspondingly, sagittal MRI sequences. The shape and any associated osteophytes of the anteroinferior acromial edge can also be seen on coronal sequences with the morphology of the bone best seen on T1-weighted images and any associated edema or inflammation noted with increased signal on T2-weighted images. Most changes seen on MRI with tendinopathy or impingement are on the bursal side of the rotator cuff in the subacromial space. There is often increased signal and hypertrophy of the bursa, as well as increased signal superficial to the rotator cuff tendons. However, with chronic tendinopathy these changes can be found within the rotator cuff muscle and tendon. In addition, one should note changes to the surrounding bony structures, including the morphology of the acromion and any changes to the AC joint (see section "Acromioclavicular Arthritis").

If there is partial tearing of a rotator cuff tendon, there is a combination of hyperintense signal from the portion of the tendon that is torn and hypointense signal from the portion that remains intact. Quantifying the exact amount of tendon that remains can be difficult. MRA can be helpful in distinguishing full-thickness from partial tears, as well as partial bursal or articular tears from intrasubstance tears.[37,38,39]

After the initial evaluation of the rotator cuff and assessment for presence of a tear, it is important to assess the size and shape of the tear, the amount (if any) of tendon retraction from the greater tuberosity, and the degree of rotator cuff atrophy. The Goutallier classification that quantifies the degree of rotator cuff muscle atrophy was based on preoperative and postoperative computed tomography scans.[3] There are four stages included in the classification as follows: stage 0, absence of intramuscular fat; stage 1, presence of some fatty streaks; stage 2, presence of substantial fat that was less than the amount of muscle; stage 3, presence of a quantity of fat approximately equal to the muscle volume; and stage 4, more fat than muscle (▶ Fig. 7.1e, f). A recent study assessing the accuracy of quantifying rotator cuff atrophy showed significant intraobserver and interobserver agreement for radiologists and orthopaedic surgeons.[40]

The medial-to-lateral width of the rotator cuff tendons (depending on the tendon) ranges from 6 to 14 mm. In addition, the tendon has an articular surface (seen within the glenohumeral joint) and a bursal surface (seen within the subacromial space). Thus, the tendon can be partially or completely torn. Assessment for the degree of tearing and location of tearing is important because it often dictates treatment. If the articular side of the tendon is torn, it is described as a partial articular-sided tear. A partial articular-sided tear of the supraspinatus is described as partial sided–articular sided supraspinatus tendon avulsion lesion. Partial tears are graded as 1, 2, or 3, according to the amount of tendon torn (grade 1 [< 3 mm, < 25%], grade 2 [3–6 mm, 25–50%], grade 3 [> 6 mm, > 50%][41]). The degree of tearing typically dictates whether the tendon is repaired or debrided.

The supraspinatus and infraspinatus are the most commonly torn rotator cuff tendons. Subscapularis tears are less common, and they are more frequently missed.[1,42] A systematic assessment of tears on multiple sequences and images allows a better understanding of various aspects of the tear. Tears of the supraspinatus and infraspinatus are best seen on sagittal and coronal T2-weighted sequences. On these sequences, if increased signal at the footprint with disruption of the normal low signal intensity of an intact tendon and/or tendons is not visualized, it indicates a tear. The degree of retraction is best seen on coronal images, whereas the anterior-to-posterior size of the tear is best measured on sagittal images. The amount of medial displacement of a torn tendon often determines whether the tear can be repaired. The tendon can remain adjacent to the normal insertion, retract medially along the articular surface of the humeral head, or retract to or past the level of the glenoid.

Tears of the subscapularis, as well as of the teres minor, are best seen on sagittal images and can be correlated with axial images (▶ Fig. 7.1g, h). Sagittal images are useful in assessing the size of a subscapularis tear, as well as determining whether the tear is on the upper or lower border of the subscapularis and whether it is a full-thickness tear. Axial images can be used to assess the degree of retraction and the presence of a subluxed or medialized LHBT. Tearing of the articular upper border is most common.[42]

Glenohumeral Arthritis

Glenohumeral arthritis typically occurs in patients older than 65 years and in men more commonly than women. In addition to the common osteoarthritis findings in other joints (hip and knee) such as joint-space narrowing, periarticular osteophytes, and cystic change, glenohumeral arthritis has distinct features

such as flattening and enlargement of the humeral head, and posterior glenoid erosion. Rotator cuff tears are uncommon, occurring in approximately 5 to 10% of patients with glenohumeral osteoarthritis.[43] However, because the presence of a rotator cuff tear can affect surgical treatment, it is a vital aspect of the preoperative assessment.

To assess the cartilage, gradient-echo sequences are the most helpful and accurate. The relationship of the articular surfaces of the glenoid and humeral head can be seen on coronal and axial images. Axial images are the most useful in the assessment of glenoid version (▶ Fig. 7.7a). Osteophytes of the humeral head and of the glenoid can be seen on coronal images. Although cysts within the humeral head (and more importantly, the glenoid) can be seen in all planes, the axial plane is most useful for assessing the specific location of the cyst within the glenoid. Precise cyst location is often important in preoperative planning of placement of the glenoid component (▶ Fig. 7.7b).

Evaluation of glenoid version is also an essential part of the preoperative plan and has traditionally been examined on axillary radiographs and computed tomography scans.[44] However, similar to the advantages discussed in the assessment of glenoid bone loss with instability, MRI has important advantages in the assessment of glenoid version (▶ Fig. 7.7c). A recent study demonstrated that MRI is reliable in assessment of glenoid version and wear, with the additional benefit of assessing the soft-tissue structures of the shoulder and the integrity of the rotator cuff.[45]

Acromioclavicular Arthritis

Arthritis of the AC joint can occur in isolation or in conjunction with other shoulder abnormalities, including rotator cuff tendinopathy and tears, as well as superior labral tears and biceps tendinopathy. Although radiographic and MRI evidence of degenerative changes are common, most patients are asymptomatic.[46] Changes in and around the joint include joint-space narrowing, hypertrophy of the joint capsule, cystic changes, and periarticular erosions. These changes are seen on all sequences that allow visualization of the AC joint, including axial, sagittal, and coronal sequences (▶ Fig. 7.8).

A study of MRI finding of the AC joint in asymptomatic patients reported that 82% of patients (41 of 50) had MRI findings consistent with AC joint arthritis.[46] In a subsequent study, the same authors compared 25 patients with symptomatic AC joint arthritis with 50 patients with asymptomatic AC joint arthritis.[47] The authors assessed which MRI findings predicted symptoms. They found varying degrees of arthritic changes in both groups, but a significantly higher grade of AC joint arthritis in the symptomatic group. In addition, the authors found that the presence of bone edema in the distal clavicle and/or acromion sides of the joint was not present in any asymptomatic patients. However, bone edema on one or both sides of the joint was present in 80% of symptomatic patients. They concluded that, on MRI, reactive bone edema is a more reliable predictor of a symptomatic AC joint than the commonly encountered degenerative changes.[47]

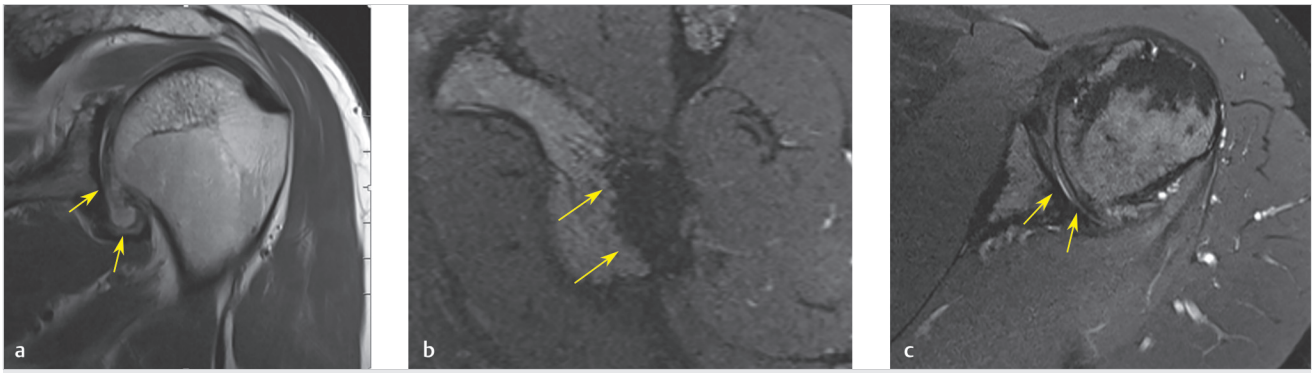


Fig. 7.7 **(a)** T1-weighted coronal image in an arthritic shoulder showing large inferior humeral head osteophyte (*yellow arrows*). **(b)** Sagittal T2-weighted image in same shoulder showing chondral wear of posterior glenoid (*yellow arrows*). **(c)** Axial T2-weighted images showing contact between humeral head and glenoid posteriorly with associated posterior glenoid wear (*yellow arrows*).

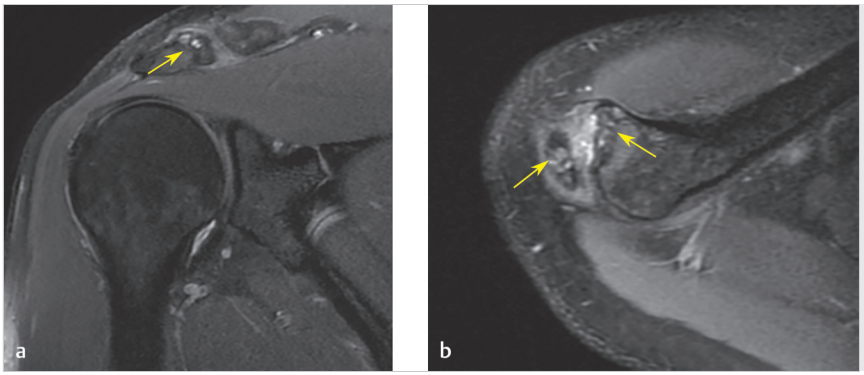


Fig. 7.8 **(a)** Coronal and **(b)** axial T2-weighted images showing acromioclavicular joint arthrosis, with joint hypertrophy and cystic change (*yellow arrows*).

7.5 Other Pathologic Conditions

7.5.1 Septic Arthritis

Septic arthritis of the shoulder is less common than septic arthritis of the hip and knee.[9,48] It can be difficult to diagnose because it often presents as vague pain that can be confused or initially misdiagnosed as a less serious subacromial bursitis, rotator cuff tendonitis, or adhesive capsulitis.[49] The confluence of symptoms can include pain (that increases with passive motion), decrease in range of motion, warmth and erythema, swelling, tenderness (typically diffuse), and fever.[49]

The work-up of a patient with possible septic arthritis of the shoulder includes radiographs, blood tests (C-reactive protein, erythrocyte sedimentation rate, and white blood cell count), laboratory analysis of joint-fluid aspirate, and a detailed history and physical examination. Patients with septic arthritis may have a history of recent shoulder surgery, injection, or illness and are typically older than 60 years. In a study of 18 patients with septic arthritis of the shoulder, 17 patients had at least 1 serious medical comorbidity, including alcoholism, cancer, diabetes, cardiac disease, cirrhosis, metabolic disease, neurologic impairment, or other infection.[49,50] Other risk factors include human immunodeficiency virus, history of intravenous drug abuse, and end-stage renal disease.[48]

Radiographs typically do not help in confirming the diagnosis of septic arthritis; however, they can aid in identifying preexisting conditions (e.g., glenohumeral arthritis, calcific tendonitis). MRI is a useful adjunct in the diagnosis and treatment of the septic shoulder. The septic shoulder typically shows a joint effusion, which can be assessed in all planes, and is best seen on T2-weighted and fat-suppressed T2-weighted MRI sequences.[51,52] Acutely, the findings that most correlate with septic arthritis are synovial enhancement, perisynovial edema, and joint effusion.[53] Postcontrast fat-suppressed T1-weighted images are helpful in evaluating for perisynovial edema and enhancement.[51] Edema within the humeral head and glenoid can also be seen in the acute stages. Chronic changes include glenoid or humeral head erosions, cartilage destruction, and with continued progression, subchondral edema and cyst formation.[51]

7.5.2 Adhesive Capsulitis

Adhesive capsulitis or "frozen shoulder" is a common condition. It classically presents in middle-aged women, diabetic patients, and patients with thyroid disorders. Patients typically present with pain that precedes progressive and global loss of glenohumeral range of motion. Unlike patients with rotator cuff tears who more typically present with decreased active range of motion and maintenance of passive range of motion, patients with adhesive capsulitis present with decreases in both active and passive range of motion, including forward elevation, abduction, and internal and external rotation.[17] Because the condition results from thickening, contracture, and inflammation of the capsule, MRI findings include thickening of the joint capsule and decreased contrast filling of the axillary pouch if MRA is obtained (▶ Fig. 7.9(a,b)).[50,54] In addition, a capsular thickness greater than 4 mm in the area of the axillary pouch, as measured on axial and coronal sequences, is often seen with this diagnosis.[55] Consistent with the physical examination finding of decreased external rotation, MRI findings can include scarring or narrowing of the rotator interval, which can be assessed on axial and coronal images.

7.5.3 Calcific Tendonitis

Similar to rotator cuff tendonitis or partial rotator cuff tears, calcific tendonitis can present as an acute or chronic painful condition of the shoulder. It can involve any of the rotator cuff tendons, most commonly the supraspinatus. Its clinical presentation is similar to that of other inflammatory conditions of the rotator cuff and includes laterally based pain that increases with overhead motion and activity, greater tuberosity tenderness, and sometimes weakness with strength testing of the rotator cuff in isolation. The intratendinous calcium deposit that characterizes the condition can be seen on conventional radiographs, typically medial to the rotator cuff footprint. MRI more clearly defines not only the specific tendon or tendons involved but also the size of the deposit (▶ Fig. 7.10). The approach to MRI review should be the same as in a patient with a rotator cuff tear, assessing the degree of involvement on coronal and sagittal images. On T1- and T2-weighted sequences, the calcium deposits are hypointense, and edema surrounding the deposits is seen as hyperintensity.

7.6 Postoperative MRI Findings

For various reasons, it is difficult to assess whether a rotator cuff tendon, labrum, or other repaired tissue has retorn after surgery. Some postoperative edema in bone and soft tissue is normal; artifact may be created by anchors or other implants; retears can be

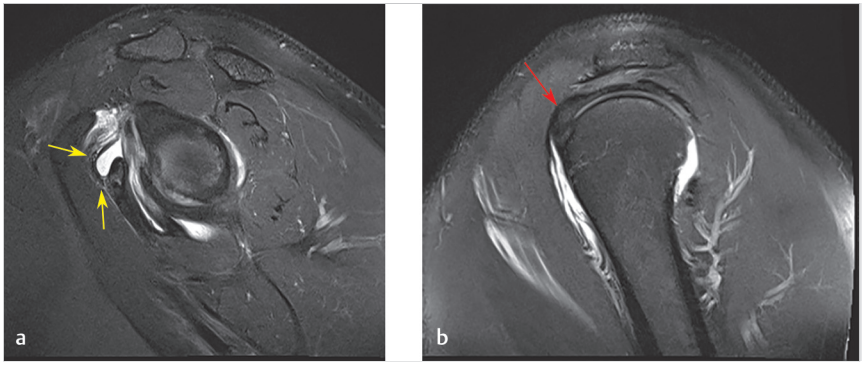


Fig. 7.9(a,b) Sagittal T1 fat saturated images post arthrography demonstrate synovial hypertrophy and synovitis seen in rotator cuff interval (*red arrow*) and subscapularis recess (*yellow arrows*).

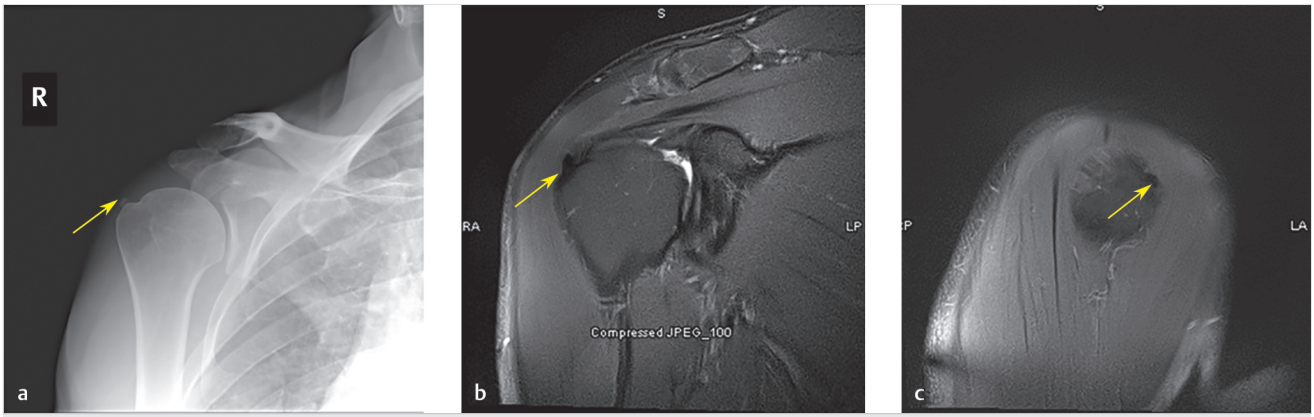


Fig. 7.10 **(a)** Plain radiograph of calcific tendonitis with typical hypointense appearance of the calcification (*yellow arrows*) on **(b)** coronal and **(c)** sagittal T2-weighted imaging.

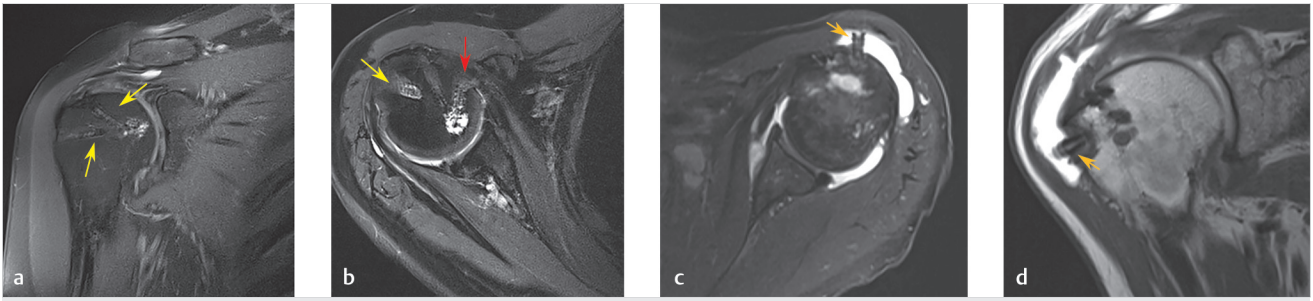


Fig. 7.11 **(a)** T2-weighted coronal image with evidence of previous double row rotator cuff repair with medial and lateral anchors in greater tuberosity (*yellow arrows*). **(b)** T2-weighted axial image showing anchors placed in greater tuberosity (*yellow anchor*) and evidence of subscapularis repair with anchors in the lesser tuberosity (*red arrow*). **(c)** Axial T2-weighted and **(d)** coronal T1-weighted images showing previous rotator cuff repair with anchor failure and pull-out (*yellow arrows*) into subacromial space.

difficult to distinguish from normal postsurgical appearance of healed tissue; and partially healed versus full-thickness rotator cuff retears can be difficult to distinguish (▶ Fig. 7.11).

There are several important tools to aid in the evaluation of the postoperative shoulder. First, one must have knowledge of the specifics of the operation performed. This allows a more accurate understanding not only of what was repaired but also where implants may have been placed and whether bone has been resected (e.g., distal clavicle excision). When available, a comparison of presurgical MRI or MRA is helpful. This allows comparison of the preoperative appearance of labral and rotator cuff tears with postoperative appearance. Challenges remain in determining the clinical relevance of an irregular appearance of a repaired tendon, including minor leakage of contrast (i.e., repair not watertight), the labrum appearing rounded but intact, as well as determining significance of fluid-like signal in the subacromial space.[56] In addition, some sequences such as spin-echo sequences tend to minimize the artifact.

Rotator cuff surgery is effective at relieving pain and, if the tendon heals, restoring function.[57,58,59] However, the retear rate after repair remains high.[58,60,61] Preoperative patient factors, tear size, number of tendons involved, amount of tendon retraction, muscle atrophy, and degree of fatty infiltration have been assessed as potential predictors of postsurgical retear. One study suggested that although MRI is sensitive for diagnosing a retear, its specificity is low. In an analysis of 37 shoulders in 33 patients with previous rotator cuff repairs who underwent revision surgery for presumed failed repair, Motamedi et al[62] found a sensitivity of 91% and a specificity of 25%. MRI correctly identified recurrent tears in 30 of 33 shoulders. However, of the four shoulders without retears, MRI was incorrectly interpreted as showing retear in three shoulders. Motamedi et al[62] concluded that, in addition to being inaccurate in diagnosing the size of retear, MRI may be associated with over diagnosis of recurrent tears in patients with symptoms consistent with retear. In this study and others, the criteria for recurrence of tear includes the absence of tendon or disruption of tendon morphology with the presence of a focal defect (tear) with high fluid signal intensity in T2-weighted imaging (coronal and sagittal views are most helpful).[16,56]

Similar to the assessment of the repaired rotator cuff tendon, the assessment of the postsurgical labrum can be challenging. Important aspects of the surgery, including repair of anterior, posterior, and superior labrum; open versus arthroscopic procedure; and use of bony augment are important to evaluate. The surgically repaired labrum loses some if its crisp, sharp edges, appearing somewhat blunted. The locations of the suture anchor tunnels, seen within the glenoid on axial, sagittal, and coronal T1-weighted images, are an obvious landmark for the site of labral repair (▶ Fig. 7.12). Thickening of the anterior or posterior capsule, depending on the site of repair, is also common.[56] However, the intact labral repair typically should not have fluid extravasation or linear high-intensity signal between the glenoid rim and labrum.

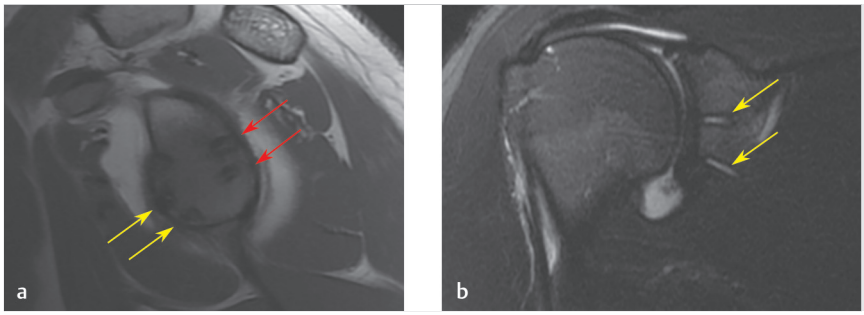
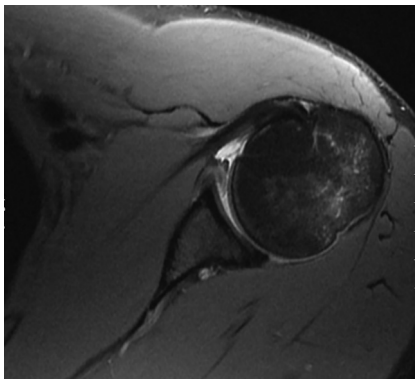


Fig. 7.12 **(a)** Sagittal T1-weighted image showing previous anterior and posterior labral repair with evidence of anchors placed anterior (*yellow arrows*) and posterior (*red arrows*). **(b)** Coronal T2-weighted image in the same patient showing anchors within the glenoid (*yellow arrows*).

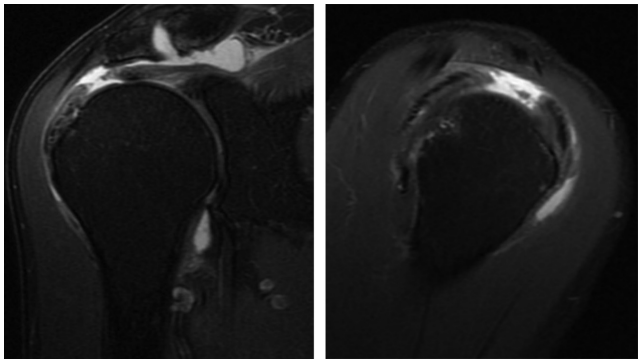
Shoulder MRI after shoulder arthroplasty, including total arthroplasty and hemiarthroplasty, is substantially affected and subject to artifact. However, MRI remains helpful in the assessment of the integrity of the rotator cuff, which can play an important role in the clinical decision-making process in the setting of a failed or painful arthroplasty. In addition, in the case of hemiarthroplasty, it can allow assessment of the glenoid articular cartilage, which is important in the assessment of a patient with hemiarthroplasty who is experiencing persistent pain. Availability and implementation of metal artifact–reducing techniques and protocols also allow an accurate assessment of not only the soft-tissue structures (rotator cuff, deltoid, capsule) but also the implant itself, including loosening and osteolysis.[63]

7.7 Common Clinical Questions

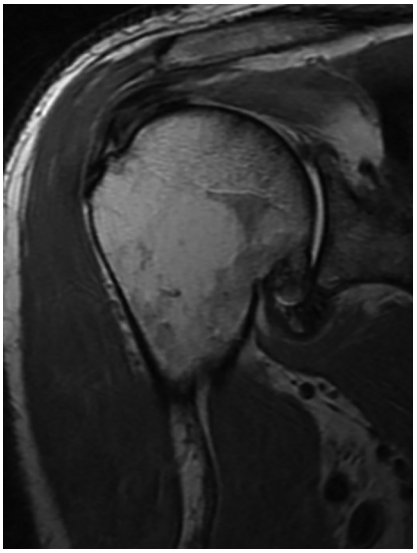
1. A 22 year old male is in your office after sustaining a shoulder injury during a football game this week. What findings are evident on the T2 weighted axial MR image below?
   a) Full thickness subscapularis tear.
   b) Dislocated biceps tendon, with upper border subscapularis tear.
   c) Posterior labral tear.
   d) Anterior labral tear, with a Hill Sachs lesion of the humeral head.
   e) Pan labral tear.



2. A 60-year-old male presents to your office with shoulder pain and overhead weakness. On examination, he has weakness in both abduction and external rotation, as well as positive provocative biceps maneuvers. Based on the T2 weight coronal and sagittal MR images below, what is his diagnosis?
   a) Full thickness supraspinatus tear.
   b) Full thickness infraspinatus tear.
   c) Full thickness subscapularis tear.
   d) Split tear of the long head of the biceps.
   e) Massive rotator cuff tear involving the teres minor



3. Based on the T1-weighted coronal MR image below, what physical examination finding would you expect based on the underlying diagnostic findings?
   a) Weakness in abduction as a result of full thickness supraspinatus tear.
   b) Pain and restricted range of motion based on glenohumeral arthritis.
   c) AC joint tenderness to palpation based on AC joint arthritis.
   d) Apprehension based on anterior labral tear.
   e) Positive liftoff based on subscapularis tear.



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Answers to Common Clinical Questions

1. Answer: d. Anterior labral tear, with a Hill Sachs lesion of the humeral head.
2. Answer: a. Full thickness supraspinatus tear.
3. Answer: b. Pain and restricted range of motion based on glenohumeral arthritis.